

ISYS90045

Professional IS Consulting

Seminar 10

Problem Solving

Dr Marc Cheong

(Based on contributions by Dr Rod Dilnutt; Christina
Parkes; James Kotsias; Dr Piya Shedden;
Chris Waddell; et al)

WARNING

This material has been reproduced and communicated to you by or on behalf of the University of Melbourne in accordance with section 113P of the *Copyright Act 1968* (**Act**).

The material in this communication may be subject to copyright under the Act.

Any further reproduction or communication of this material by you may be the subject of copyright protection under the Act.

Do not remove this notice

Agenda:

Part 1. Problem Solving (Marc)

Intended Learning Outcomes (ILO's)

- Define and understand structured problem solving
- Define and understand unstructured problem solving
- Engage in activities and examples to gain understanding

Part 2. Assignment 2 Briefing (Christina)

Special thanks & credits
to James Kotsias
for designing the initial
version of this seminar.

Delivery and unstructured
problem solving

James Kotsias

Introduction to Problem Solving

Consultants solve problems.

IT, IS and information security consultants solve problems specific to their domain:

Should we go to the cloud?

How can we improve our security environment?

How does this software best support the business?

What is our IT roadmap?

How effective is our data loss prevention?

We want to go with SAP - can you make it happen?

We're outsourcing IT - who should we go with?

We have major IT uplift going on - please manage it!

Do any of these scenarios have easy answers?
How do you even begin solving these?

Defining Problem Solving

Problem solving

- It is an analytical process used to identify the possible solutions to the situation at hand.
- Making decisions is a part of problem solving.
- Problem solving is a complex process, and judgement calls (or decisions) will have to be made on the way.



Exercise:

You will be presented with a modern problem for discussion... relating to Generative AI!

Introduction to problems and decisions

Types of problems we encounter:

- Structured
- Unstructured
- (and Semi-structured)

How does this incorporate decision making?

- Programmed decisions
 - e.g., Standard Operating Procedure (SOP) to eliminate individual judgment
- Non-Programmed decision
 - e.g., Strategy to determine whether to acquire another company

Introduction to problems and decisions

Think of problem solving in two ways:

Structured problem solving

- Problems that are known, standardised and can be solved in an operational fashion. Can be encoded into procedure documentation.
- For example:
 - How do you make macaroni and cheese?
 - How do you resolve customer enquiries in a call centre?

Unstructured problem solving

- Problems that are open-ended, 'fuzzy', may be unique to a situation
- For example:
 - How is macaroni better than fettucine?
 - What should we do with our IT systems in the next three years?
 - Should we acquire a new business?
- How do you solve these?

Problem solving: structured

‘Well-structured’, clear definition of the:

- Goal
- Beginning state
- Actions
- End state
- Constraints

Follow similar processes for resolution

- Procedural; limited number of rules and principles
- Method for solving the problem is repeatable
- Outcomes are typically known or standardised
- Constraints are limited or are accounted for.

(Jonassen, 1997)

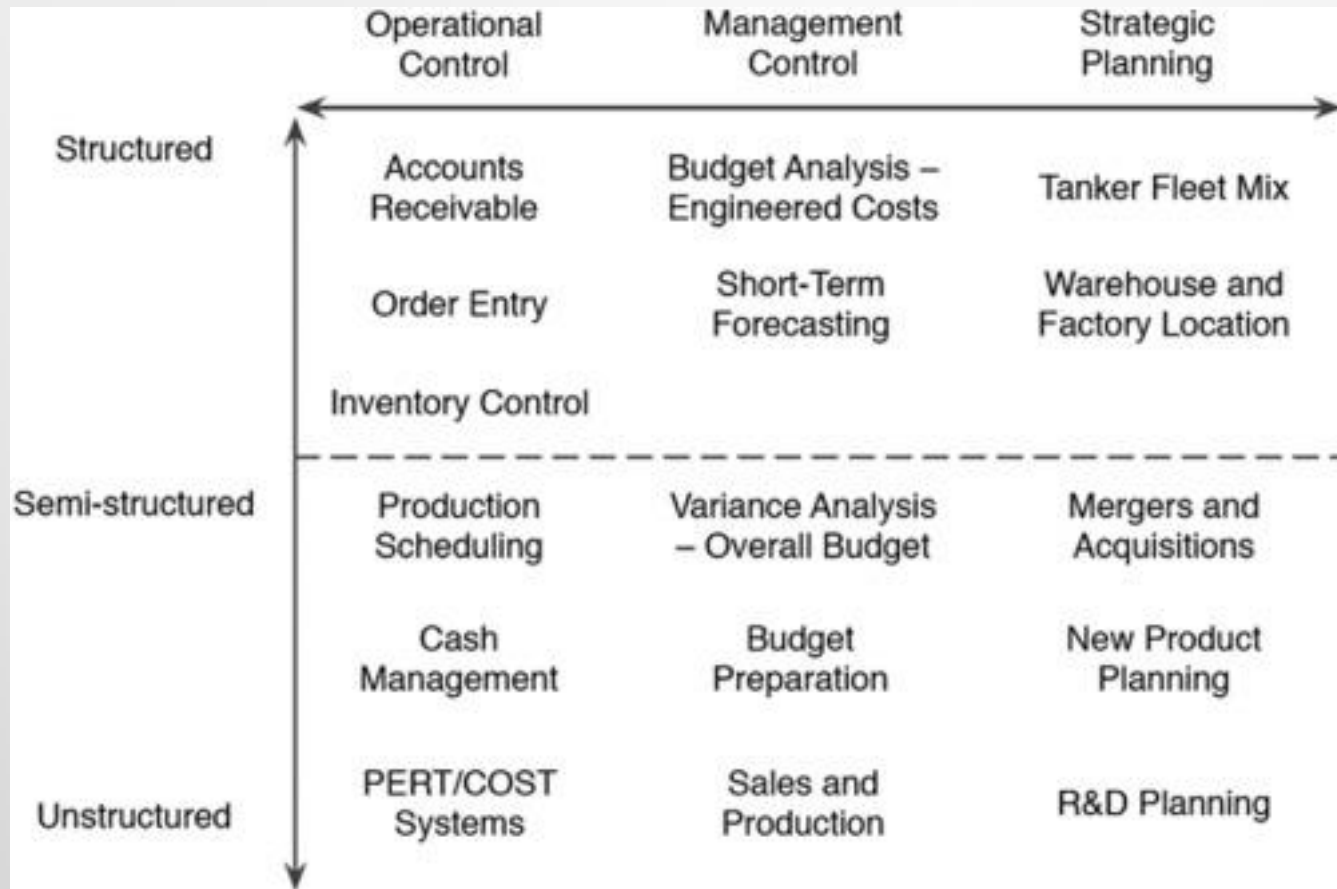
Problem solving: unstructured

‘Ill-structured’ problems

- Unknown definition of the problem elements
- Multiple potential solutions
- Different paths to take to get to the solution
- Uncertainty over concepts, rules to apply
- Not enough information on hand
- Learners need to make judgements on the problem – and defend this view

(Jonassen, 1997)

Problem solving: unstructured



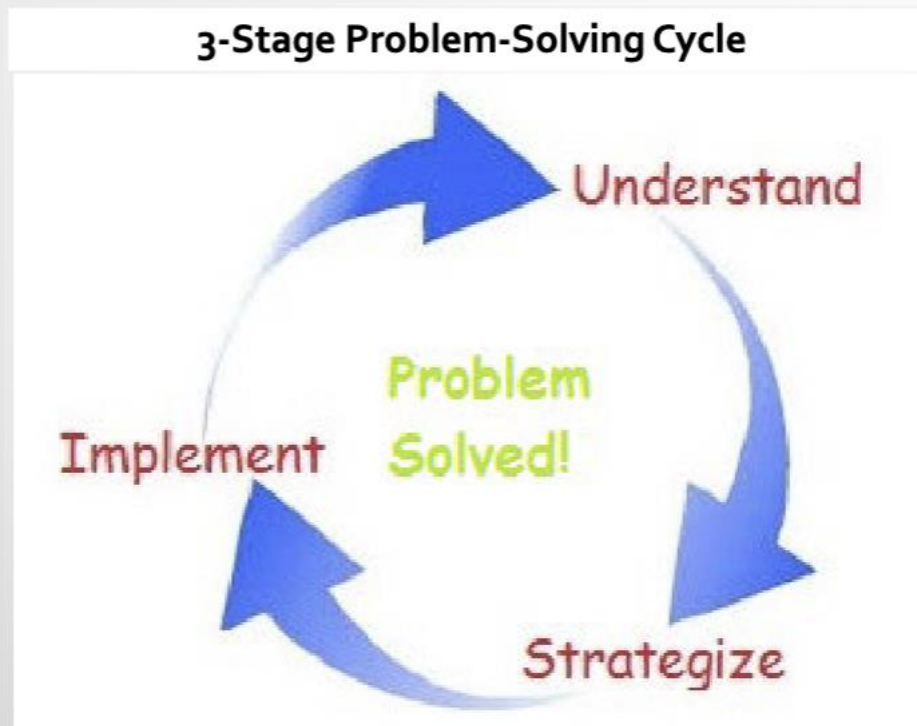
Source: (Gorry and Scott Morton, 2001;
as presented in Arnott, Lizama, and Song, 2017).

Comparison: (Well)structured vs Unstructured

Characteristics	(Well-)Structured problems	Unstructured / ill-structured problems
Decision type?	Programmed	Nonprogrammed
Frequency	Repetitive/routine	New/unusual
Goals	Clear/specific	Vague
Information	Readily available	N/A or unclear
Consequences	Minor	Major
Organisational level	Lower	Upper
Time for solution	Short	(Relatively) long
Basis for solution	Decision rules/set procedures/SOP	Judgement/creativity
Specifications	Unambiguous/complete	Ambiguous/incomplete
Evaluations	Defined evaluation criteria/process	No definite criteria/stopping rule

Adapted from (Griffin and Moorhead 2014, p. 211/Table 8.1);
(Gasson, 2006; citing Guindon, 1990).

Problem solving – An Approach



IMPORTANT: there are several models (such as the McKinsey 7-Step) – but for this subject and assessments, we will focus on the 3-stage cycle.

Source:
(North Carolina School of Science and Mathematics, 2016).

The Three Stages Explained:

#1. Understanding the Problem

The most important part of solving any problem is understanding it!

Ask yourself or guide others to ask themselves the following questions:

- What am I looking for?
- What is the unknown?
- Do I understand every word and concept in the problem?
- Am I familiar with the units in which measurements are given?
- Is there information that seems missing?
- Is there information that seems superfluous?
- Is the source of information bona fide?



Exercise: Misunderstanding the Problem

**Anecdote – how a problem is misunderstood;
and hence, unsolvable!**

Source: “DaveHax” (2016). The 3 Cups Challenge - You always win the bet!.

YouTube. https://www.youtube.com/watch?v=qxqPZ_K3uig



The Three Stages Explained:

#2. Strategising

Now that we think we understand the problem, we choose a strategy or a set of strategies to try to solve the problem.

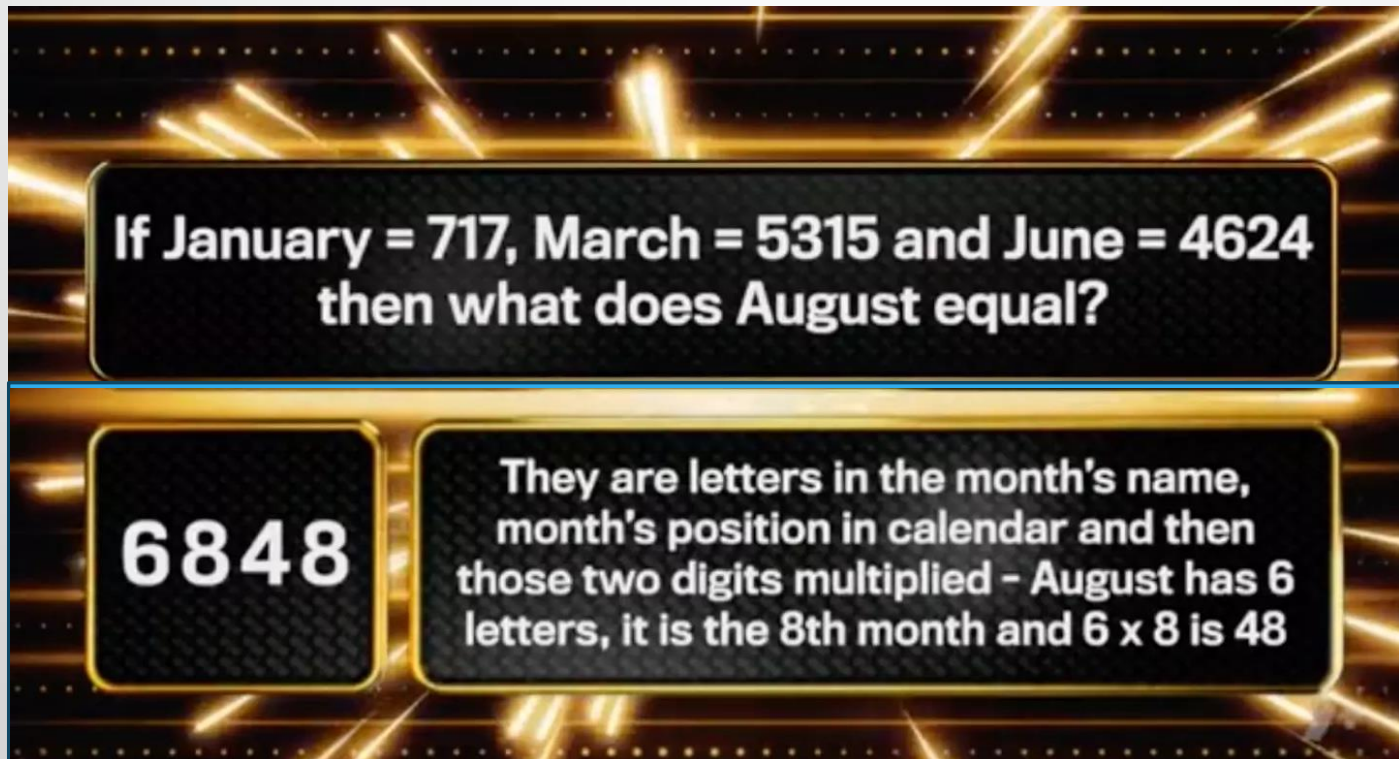
Ten general strategies are:

1. Logical reasoning
2. Pattern recognition
3. Working backwards
4. Adopting a different point of view
5. Considering extreme cases
6. Solving a simpler analogous problem
7. Organizing data
8. Making a visual representation
9. Accounting for all possibilities
10. Intelligent guessing and testing

Source: (North Carolina School of Science and Mathematics, 2016; citing Posamentier and Krulik, 2015).

▶ Exercise: Strategies in Action

Problems are from the TV show “The 1% Club” Australia.
(Leahy, 2023).
Image via Tiktok.



**If January = 717, March = 5315 and June = 4624
then what does August equal?**

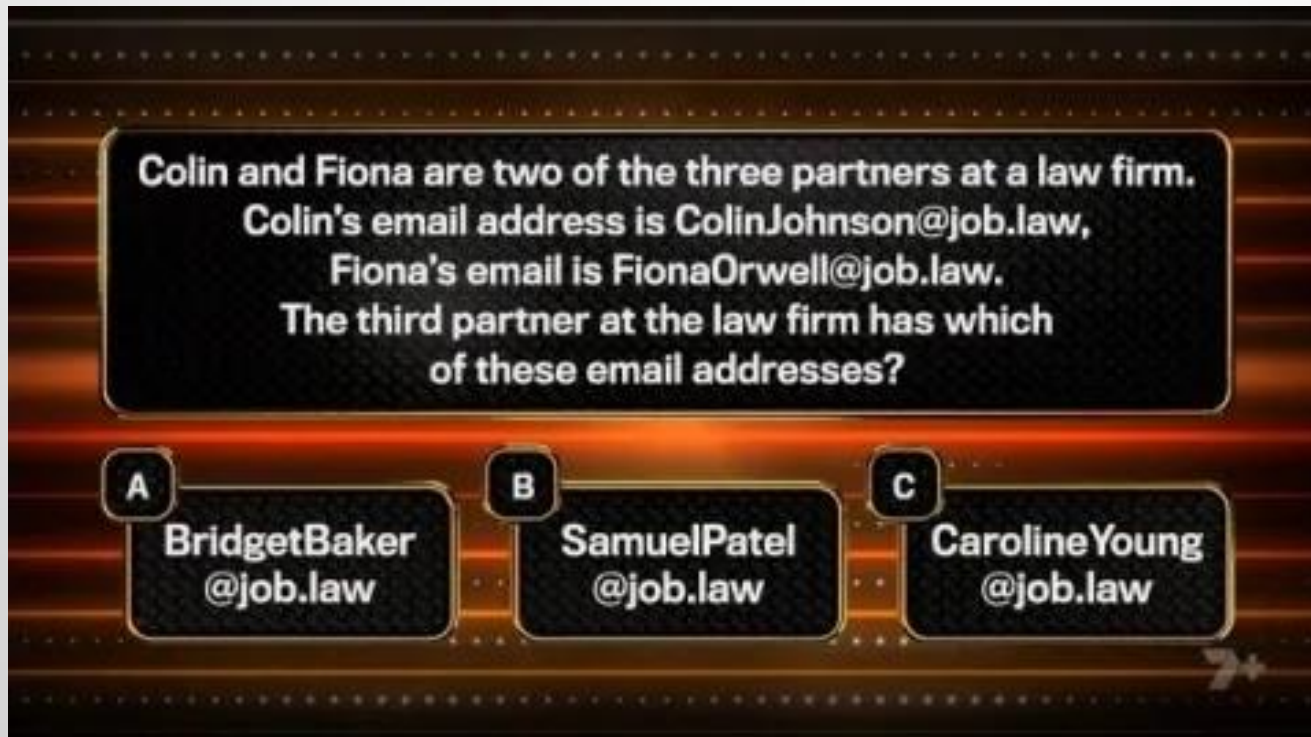
6848

**They are letters in the month's name,
month's position in calendar and then
those two digits multiplied - August has 6
letters, it is the 8th month and 6×8 is 48**

▶ Exercise: Strategies in Action

Problems are from the TV show “The 1% Club” Australia.
(Leahy, 2023).

Image via Yahoo Lifestyle. <https://au.lifestyle.yahoo.com/the-1-club-how-many-of-these-questions-from-sevens-new-game-show-can-you-solve-233819307.html>



The Three Stages Explained:

#3. Implementing

- Implement the strategy or set of strategies
 - Check reasoning and calculations.
- Avoid the mistake of “doing something” **before understanding** or at least assuming understanding.
 - Note: incorrect understanding != not realizing that we need to understand.
- As we implement we might refine our understanding of the problem.
 - As we refine our understanding of the problem →
 - As we refine our strategy →
 - As we implement a new approach...
 - ...we get closer to solving the problem, etc

Problem Solving Presentation Principles I

In business we constantly look at ways to solve prospects', clients' and customers' problems. When delivering proposal presentations consider:

The Problem

- Clearly describe the problem that the customer has.
- Ensure that they agree that this is a problem and that it needs solving.
- May be necessary to show or tell them the implications of not solving the problem and highlight the pain caused.

Example: A car sales person asks why the customers are looking for a new car. The customer tells how the current car is expensive to run...

Problem Solving Presentation Principles II

In business we constantly look at ways to solve prospects', clients' and customers' problems. When delivering proposal presentations consider:

The cause

- Show the main causes of the problem
- Make links between causes and the problem very clear
- Use a maximum of three causes.

Example: A car sales person asks why the customers are looking for a new car. The customer tells how the current car is expensive to run...
...The salesperson explains how some cars are not designed for easy service and others have inefficient engines.

Problem Solving Presentation Principles III

In business we constantly look at ways to solve prospects', clients' and customers' problems. When delivering proposal presentations consider:

The solution

- Show your product/service addresses the causes and fixes the problem
- Show independent reports that prove this.
- Demonstrate if possible.

Example: A car sales person asks why the customers are looking for a new car. The customer tells how the current car is expensive to run...
...The salesperson explains how some cars are not designed for easy service and others have inefficient engines
...She then shows them a report that highlights running costs of different cars and then demonstrates models from the top two cars in the list.

Problem solving: unstructured

Unstructured problem solving requires **structure**.

You need to **break the problem down**.

The below model illustrates an effective means of logically stepping through this process. (Rosenberg, 2011).



Problem definition: What is the nature of the problem, root cause or goal? Consider your scope and size of application. Alternative mechanisms and paths if ideal tools are not available.

Criteria selection: How will you distinguish between alternatives? Qualitative vs Quantitative? Weighting? Risks and downsides?

Options: What are the alternative methods for solving the problem?

Problem solving: unstructured

Unstructured problem solving requires **structure**.

You need to **break the problem down**.

The below model illustrates an effective means of logically stepping through this process. (Rosenberg, 2011).



Analysis: Apply your criteria and distinguish between the alternatives. If necessary, think of mitigations for risks that you've considered.

Decision: Apply your criteria and distinguish between the alternatives.

Action: Develop the plan to implement the decision.

IMPORTANT: there are several unstructured problem-solving models – but for this subject and assessments, we will focus on the Rosenberg (2011) model.

Jonassen (1997): Attributes of Problems

Instructional Design Models for Well-Structured
and Ill-Structured Problem-Solving Learning
Outcomes

□ David H. Jonassen

- **Domain:** content that outlines the problem's element
 - Concepts, rules, principles -- e.g., information systems
- **Type:** A combination of related concepts and rules that will help solve the problem.
 - e.g., information security risk management
- **Process:** Method by which you solve the problem
 - Depends on your understanding of the problem type, goal.
 - Gap between initial and goal state is the problem 'space'
 - This can become more proceduralised as you gain experience
- **Solution:** Achieving the goals
 - Convergent vs. divergent (one solution vs. one of many)

Tame vs Wicked Problems!

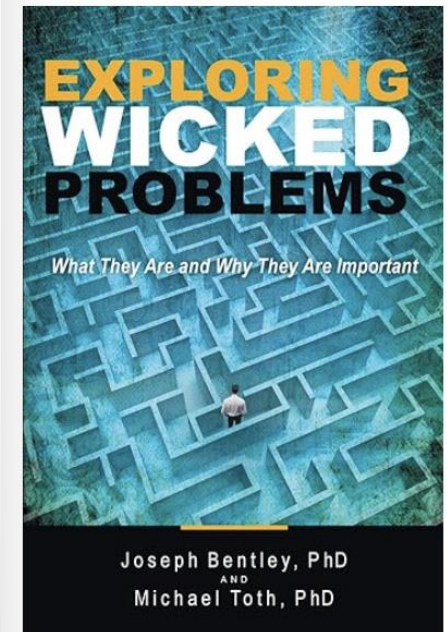
‘Former Secretary of State George Shultz once drew a distinction between "**problems you can solve** and **problems you can only work at**"...’
(Bentley, 2014).

* Bentley co-wrote a whole book on this →

▶ Exercise:

What are examples of ‘tame’ problems?

What are examples of ‘wicked’ problems?



10 Properties of Wicked Problems (Rittel and Webber, 1973)

Example applied to the wicked problem 'poverty' (Rittel and Webber, 1973)

Policy Sciences 4 (1973), 155–169
© Elsevier Scientific Publishing Company, Amsterdam—Printed in Scotland

Dilemmas in a General Theory of Planning*

HORST W. J. RITTEL

Professor of the Science of Design, University of California, Berkeley

MELVIN M. WEBBER

Professor of City Planning, University of California, Berkeley

1. There is no definitive **formulation** of a wicked problem
2. Wicked problems have no **stopping rule**
3. Solutions to wicked problems are **not true-or-false**, but **good-or-bad**
4. There is no immediate and **no ultimate test of a solution** to a wicked problem
5. Every solution to a wicked problem is a "one-shot operation" [...] **no opportunity to learn by trial-and-error**

10 Properties of Wicked Problems (Rittel and Webber, 1973)

Example applied to the wicked problem 'poverty' (Rittel and Webber, 1973)

Policy Sciences 4 (1973), 155–169
© Elsevier Scientific Publishing Company, Amsterdam—Printed in Scotland

Dilemmas in a General Theory of Planning*

HORST W. J. RITTEL

Professor of the Science of Design, University of California, Berkeley

MELVIN M. WEBBER

Professor of City Planning, University of California, Berkeley

6. Wicked problems **do not have an enumerable** (or an exhaustively describable) **set** of potential solutions[...]
7. Every wicked problem is essentially **unique**
8. Every wicked problem can be considered to be a **symptom** of another problem
9. The existence of a discrepancy representing a wicked problem can be **explained in numerous ways**. [...]
10. The planner has **no right to be wrong**

Industry Perspectives: Leadership and Problem-Solving Skills



Harvard Business School (Cote, 2003)

“Four Problem-Solving Skills All Leaders Need”

1. Problem Framing
2. Empathy
3. Breaking Cognitive Fixedness
4. Creating a Psychologically Safe Environment

Industry Perspectives: Mindsets

Faced with an inscrutable problem? Try a more creative approach.



McKinsey Quarterly

Six problem-solving mindsets for very uncertain times

September 15, 2020 | Article

McKinsey
& Company

Problem solving and Consultants

‘Collective cognition’ in problem solving

- Distributed cognition and creativity
- Groups of individuals interacting to layer meaning, perspectives and experiences to solve problems (Hargadon & Bechky, 2006)

Consultants are more creative and innovative?

- “Consultants were found to be, on average, more cognitively inclined to produce creative and original ideas and function effectively in unstructured settings” (Summers, Sweeney & Wolk, 2000).

Diversity in consulting teams

- harness different experiences of members
 - reduces confirmation bias and groupthink
- (Source: ISYS90032 lectures by Suelette Dreyfus/Charlynn Miller; FIT4005 lectures by David Green).

Summary for Week 10

It is critical to use a structured approach to solve problems.

If the wrong problem is chosen

... the correct root causes are not identified

... the solution chosen will not improve the situation

and the problem will recur.

The foundation of Problem Solving is to spend time to
properly identify and define the problem.

“Never Assume”

“Solve it Right First Time”

References (in order of appearance)

- Jonassen, D.H. (1997). Instructional Design Models for Well-Structured and Ill-Structured Problem-Solving Learning Outcomes, *ETR&D*, 45:1.
- Arnott, D., Lizama, F., & Song, Y. (2017). Patterns of business intelligence systems use in organizations. *Decision Support Systems*, 97, 58–68.
- Gorry, G.A. and Scott Morton, M.S. (1971). A Framework for Management Information Systems. *Sloan Management Review*.
- Griffin R. W. & Moorhead G. (2014). Organizational behavior : managing people and organizations (11th ed.). South-Western/Cengage Learning.
- Gasson, S. (2006). Emergence in Organizational 'Problem-solving': Theories of Social Cognition (Last updated 10/04/2007 12:09:36) Working Paper. retrieved 26 Mar 2024. Available from <http://cci.drexel.edu/faculty/gasson/papers/probsolv.pdf>
- Guindon, R. (1990). Knowledge Exploited By Experts During Software System Design. *International Journal of Man-Machine Studies*, 33. October 1990. pp 279-304.
- North Carolina School of Science and Mathematics. (2016). The 3-Step Problem Solving Cycle. YouTube. <https://www.youtube.com/watch?v=3FogEQyO6oI>
- “DaveHax” (2016). The 3 Cups Challenge - You always win the bet!. YouTube. https://www.youtube.com/watch?v=qxqPZ_K3uig
- Posamentier, A. S., Krulik, S. (2015). Problem-solving Strategies In Mathematics: From Common Approaches To Exemplary Strategies. Singapore: World Scientific Publishing Company.

References (in order of appearance)

Leahy, J. (Executive Producer). (2023 - present). *The 1% Club Australia* [TV series]. BBC Studios Australia and New Zealand; Seven Network.

Rosenberg, M. (2011). IESE's Six Step Process for Resolving Unstructured Problems. Available from <http://blog.iese.edu/rosenberg/files/2011/12/Six-Step-Process-for-Resolving-Unstructured-Problems.pdf>

Bentley, J. C. (2014). *About wicked problems*. Taming Wicked Problems. Available from <https://tamingwickedproblems.com/wicked-problems/>

Horst W. J. Rittel, & Webber, M. M. (1973). Dilemmas in a General Theory of Planning. *Policy Sciences*, 4(2), 155–169. <http://www.jstor.org/stable/4531523>

Cote, C. (2023, January 17). *Why problem-solving skills are essential for leaders*. Harvard Business School - Business Insights Blog. <https://online.hbs.edu/blog/post/problem-solving-in-business>

Conn, C., & McLean, R. (2020, September 15). *Six problem-solving mindsets for very uncertain times*. McKinsey & Company. <https://www.mckinsey.com/capabilities/strategy-and-corporate-finance/our-insights/six-problem-solving-mindsets-for-very-uncertain-times>

Hargadon, A.B. and Bechky, B.A. (2006). When Collections of Creatives Become Creative Collectives, *Organization Science*.16:4.

Summers, S.L., Sweeney, J.T. and Wolk, C.M. (2000). Problem-Solving Style and Fit in Consulting and Auditing'. *Journal of Information Systems*. 14:1.